Real MVP: The Role of Coaches in NBA Success

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Abstract

The degree to which coaching impacts the outcome of a basketball season is much debated among armchair analysts and anecdotally by those in the sports business, but has been subject to little systematic study. Using data going back to the 1970s, this paper explores the high frequency of front office changes in the National Basketball Association (NBA) to disentangle the relative contribution of coaches in light of roster characteristics. It finds that the fixed effects for some coaches, like Greg Popovich, are larger than superstars on All-star seasons. By regressing on statistics from their playing careers, it also investigates observable characteristics that shape effective coaches.

1 Introduction

Researchers have long articulated the importance of managers and their role in elevating team performance. According to leadership experts, effective managers display strong emotional intelligence, are able to nurture talent, and are not afraid to break the rules. But while managers are at the forefront of corporations, the bulk of sports-related research has focused on players as the primary success factor, leaving coaches as an arbitrary element. This is backdropped against tremendous personal achievements that put to question their sidelining. For example, Phil Jackson has won more championships (13) than all but two NBA franchises, Gregg Popovich's long-standing success streak has reached 21 consecutive post-season appearances, and Steve Kerr has achieved consecutive finals appearances in each of his first five coaching seasons.

Despite advancements in analytics, the basketball world continues to lack any systematic quantification of the coaching effect. Traditionally, those with most silverware are considered all-time greats. But while team success is measurable, the degree to which coaches contribute to it remains a matter of conjecture. This paper overcomes this challenge by advancing a robust empirical framework for controlling for roster quality and disentangling the managerial contribution of coaches.

Coaches influence the outcome of a basketball seasons in two ways: First, they set strategy (i.e. they design and implement schemes that maximize on their players' relative strengths and expose opponents' weaknesses). Second, they inspire and establish a culture. Some of the most decorated coaches in NBA history are known for their interpersonal skills. Famous examples include Pat Riley and Phil Jackson who display an unmatched ability to spread a culture of winning throughout their organization. This paper views coaching as a simulatenous combination of both of these channels.

Following the literature on managerial contributions in organizational settings, this paper adopts a coach-fixed effect methodology. Using data from the National Basketball Association (NBA), it estimates OLS regressions for 30 teams and 165 coaches across a time period of 38 seasons (1980 to 2017). This approach follows [1] Abowd et al. (1999) who look at employee wages

in corporate sectors, [6] Bertrand and Schoar (2003) who investigate the impact of CEO identity on balance sheet performance, and [15] Muehlheusser et al. (2016) who undertake a similar analysis on coaches in professional European soccer. The purpose is to identify the degree to which various NBA coaches are able to elevate player performance in light of roster characteristics.

Some studies have looked at the collective impact of coaches but few have analyzed them at the individual level. Of those that explore the topic, the findings are mixed. Early work by [9] Gamson and Scotch (1964) finds that coaches in Major League Baseball are inter-changeable. More nullifying conclusions were reached by [7] Cannella and Rowe (1995), [8] Fabianic (1994), and [16] Smart and Wolfe (2008). In the context of European soccer, [2] Audas and Goddard (2008) find that coaching changes have a harmful effect and [15] Muehlheusser et al. (2016) find a wide degree of heterogeneous effects. In the context of the NBA, [4] Berri et al. (2009) find that coach coefficients are indistinguishable from one another; and [10] Giambatista (2004) finds detrimental effects associated with coaching changes. More recently, [**?**] Berry and Fowler (2019) examine the aggregate effect of coaches and find that some sports are more conducive to coaching than others.

The basketball world offers considerable advantages for the question at hand. First, it is unique in its high frequency of front office changes. Around 60 percent of coaches in the sample are observed on multiple teams. This helps trace their impact as they move between different sets of players. Second, the availability of advanced metrics makes it possible to capture player ability accurately and comprehensively. Finally, longitudinal data allows to match coaches' contributions with statistics from their playing careers, allowing for an investigation of observable characteristics that make for great leaders.

The findings reveal that the extent to which coaches play a role in team success is larger than that of players on All-Star caliber seasons. While some are associated with a negligeable effect, others are able to elevate player performance to new heights. For instance, changing from the mean coach (Mike Dunleavy) to the top coach (Larry Bird) generates an additional 14.2 wins when roster characteristics are held at their means. This amount is sufficient to push a mediocre team into playoff contention. Compared to a standard approach of looking at career win percentages, these fixed effect estimastes are able to take into account the quality of the roster.

Besides being interesting in their own right, these derived measures of coaching ability are helpful in predicting team performance for future coach-team pairings. In order to analyze their out-of-sample performance, this paper considers a leave-one-out cross validation approach whereby information from a coach's all but one tenures are used to predict his performance in the tenure that is left out. The results suggest that these estimates add significant power to a standard forecasting approach of looking at roster quality alone.

The paper is structured as follows. Section 2 lays out the empirical framework, including a description of the dataset, the computation of the roster quality variable, and model specification. Section 3 displays the main results in terms of aggregate and individual coach fixed effects. In Section 4, there are two main robustness checks. First, a cross-validation approach highlights the predictive power of the fixed effect estimates in forecasting out-of-sample performance for newly assigned coach-team pairs. Second, the use of performance-based metrics is addressed by considering different tenure length limitations. Section 5 uses the estimated fixed effects as a dependent variable and looks into potential correlations with background statistics from coaches' playing careers. Finally, conclusions are drawn in Section 6.

2 Empirical Framework

2.1 Data

The NBA is comprised of 30 teams that compete over 82 games each season. Performance during the regular season determines their ranking going into the post-season, which in turn affects their championship hopes. Those with the highest total wins gain home court advantage and compete against the lowest seeds. The NBA season is structured in a way that each team faces every other team at least once in a season, so the difficulty of the schedule is the same every year.

The data set covers all regular seasons played between the 1978-79 and 2016-17 seasons. In total, this comes down to 1,063 individual team-seasons and includes 165 individual head coaches. The unit of analysis is the performance of a coach-team pair over the course of a regular season

and the dependent variable is the total number of wins. There are two lockout seasons that are adjusted to fit the rest of the sample. All the data is made available by Basketball Reference.

Each coach is matched with performance metrics from his players' past performances. During seasons where multiple coaches are observed on the same team, only the coach with the highest number of games is considered for analysis. As mentioned previously, the rate of coaching changes in the NBA is exceptionally high: 63 percent of coaches in the data set are observed on multiple teams. The average career duration for an NBA coach extends to 7.1 seasons and spans 2.4 teams (Table 1). Figure 1 displays career win percentages in function of the number of coaching years. Table A2 (Appendix) provides detailed information on all the coaches in the sample.

Table 1: Summary Statistics Per Head Coach

	Mean	Std. Dev.	Min.	Max.
Seasons	7.1	6.6	0.9	30.7
Teams	2.4	1.6	1	11
Games	577.2	537.6	71	2487
Wins	291.9	309.3	11	1335

Time period: 1953-54 until 2016-17 season. Note: Includes 165 coaches, excluding those with less than one full year of coaching experience.

2.2 Model Specification

The main estimation strategy is a coach-fixed effect (FE) regression model with team wins as a dependent variable. As independent variables, the model includes a dummy variable representing head coach identity and control variable for roster quality.

The following empirical model explains team i's wins under coach k's in season t:

$$\text{Wins}_{itk} = \beta_1 \text{RosterQuality}_{i,t} + \lambda_k + \epsilon_{itk} \tag{1}$$

where $\text{RosterQuality}_{i,t}$ represents the team's estimated talent level using previous year performance metrics, coach-fixed effects are captured by λ_k, and ϵ_{itk} is the error term.

The Roster Quality variable is computed using performance metrics from the previous

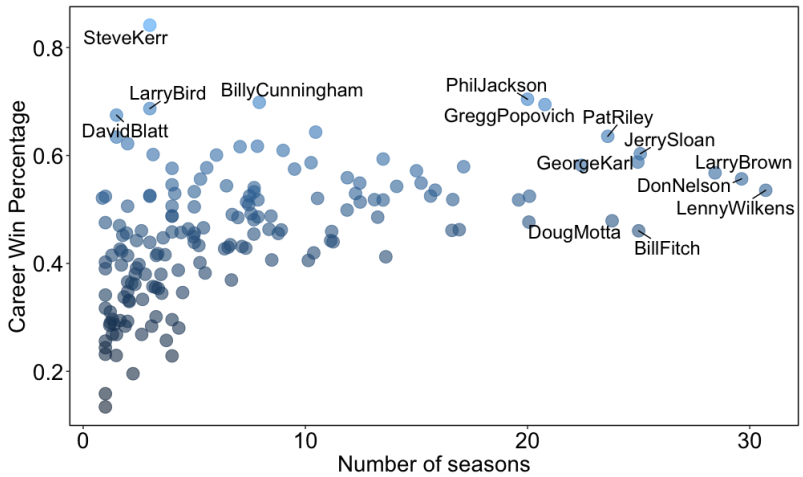


Figure 1: Career Performance of NBA Coaches

season. More specifically, it uses Box-Plus Minus (BPM) as the metric of choice . For each team, a minute-weighted average is computed using BPM from the season prior. Weighing by total minutes played serves to account for players that are relied upon more heavily than others. Figure 2 shows the relationship between estimated roster quality and number of wins, where there seems to be a positive correlation.

It is useful to note that basketball coaches do not usually make hiring decisions, which means they have no control in choosing their roster. This is typically the responsibility of the General Manager, who may have different considerations for signing certain players. Only in a limited number of cases has the coach been also assigned managerial powers. In this regard, basketball sets itself apart from soccer or the NFL, for example, where the coaching role coincides with franchise decisions. This distinction is important because the roster variable needs to be independent from the coaching variable for the regression model to accurately capture the coaching effect.

There are meaningful differences across teams that are not captured by player metrics. Things like front office quality, training facilities, medical staff, and geographical location could influence performance. Big market teams such as in Los Angeles, New York, and Boston are known to have more traction in the free agent market, which could make them better-suited for success. One way to account for such differences is through team fixed effects. However, there is enough reason to believe that team fixed effects should not be considered in the main specification. The

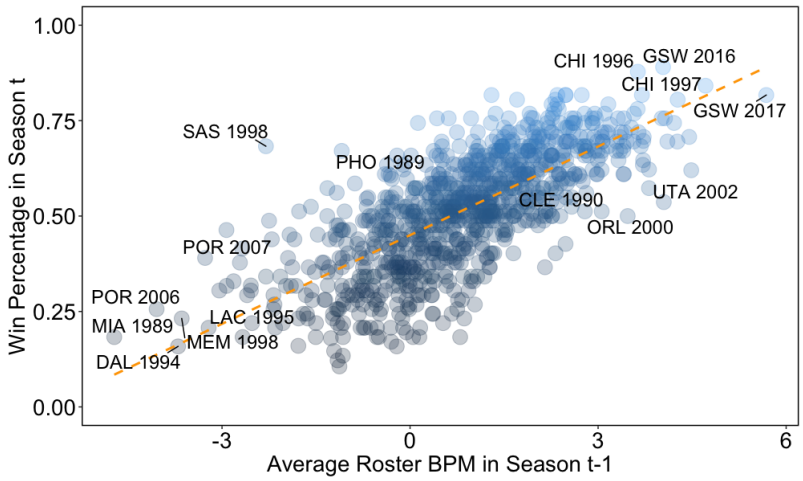


Figure 2: Estimated Roster Quality Variable versus Team Performance

nature of the NBA is such that losing teams are compensated through the draft, which ensures a well-balanced structure of power, where no team possesses a systematic advantage.

A fixed effect methodology is particularly suitable for environments where there is a frequent and pseudo-random allocation of managers. For example, Bernard and Schoar (2003) [6] employ a fixed effect approach to investigate whether the identity of CEOs matters for a companies' financial performance. Abowd et al. (1999) [1] use person effects to explain wage variation at French firms. Beaudry and Willems (2018) [3] apply this framework to examine the effect of managerial tendencies by IMF Mission Chiefs on countries' macroeconomic fluctuations. And most relevant to the context of this paper, Muehlheusser et al. (2016) [15] undertake a similar analysis on coaches in professional German soccer.

3 Results

This section investigates whether head coach identity matters for team performance. It involves two exercises: Following Muelheusser et. Al (2016) [15], I start by comparing the joint impact of coaches against a benchmark model that does not include coach fixed effects. The purpose is to determine whether the explanatory power of the regression increases once individual fixed effects are included. As a second step, I analyze the individual fixed effect associated with each head coach in order to identify those who tend to elevate player performance more than others.

3.1 Joint Impact of Coaches

Table 2 displays the results from the main model, which includes coach fixed effects, versus a benchmark model that only contains roster quality as independent variable. While both models point to a high correlation between roster quality and team wins, the explanatory power is considerably higher in the main model. The adjusted R-squared increase by 10.9 percentage points and the joint significance F-test on the coach fixed effect is highly statistically significant ($p < 0.01$).

Table 2: Joint Impact of Coaches

	Dependent variable: *Wins*	
	Benchmark Model	Main Model
Roster Quality	*Yes*	*Yes*
Coach Fixed Effects	*Yes*	*Yes*
N	1,063	1,063
Adj. R-squared (%)	48.3	59.2
F-Test Coach FE (p-value)	-	0.000

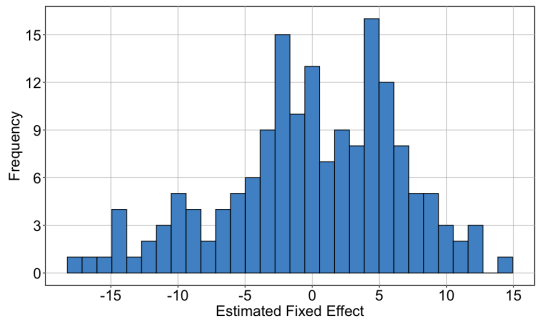
Time period: 1978-79 until 2016-17 season. Note: Standard errors are clustered at the team level.

3.2 Individual Coach-Fixed Effects

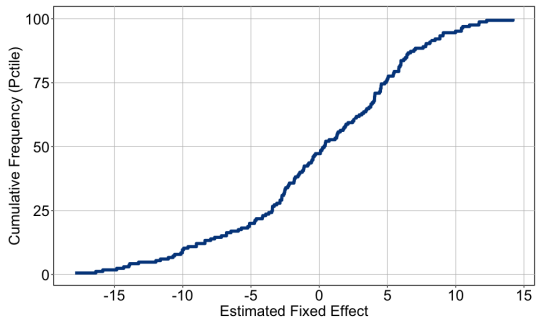
The fixed effect methodology estimates the average impact associated with the identity of every NBA head coach. With team wins as a dependent variable, the estimated fixed effect comes in the form of the number of additional wins contributed, on average, each season. While coaches with tenures on multiple teams are associated with lower standard errors, fixed effects can also be identified for non-movers by disentangling them from the effect of roster quality.

Figure 3 displays the estimated fixed effects for the top 20th percentile of coaches. The full list of estimates along with statistical significance can be found in Table A1 in the Appendix, where a ranking by fixed effects is compared against a more naïve ranking using career win percentage. The fixed effects are demeaned, so they represent average additional wins relative to the mean head coach (here, Mike Dunleavy).

Overall, the fixed effects are quite large for some coaches - enough to propel mediocre



(a) Frequency Distribution



(b) Cumulative Distribution

Figure 3: Frequency Distributions of the Estimated Fixed Effects

teams into playoff territory. For example, changing from the mean head coach to Steve Kerr generates 12.3 additional wins, on average, when roster characteristics are held at their means. To put this into perspective, this amount is greater in magnitude than all-star Damian Lillard's win share metric during the 2018-19 season (where he ranked 4th overall). This suggests that, contrary to mainstream belief, coaches are at the forefront of team performance, with contributions that are at times more valuable than the team's best player. With Kerr's fixed effect, for example, a team ranked 11th by conference standing during 2018-19 would have been able to finish sixth overall.

Coaches with fewer observations are subject to higher standard errors associated with the estimated fixed effect. For instance, Bird (1), Kerr (2), and Ainge (10), all of whom figure in the top 20th percentile of coaches, are limited to three observations each. This is due to their relatively short tenure as head coaches. Both Bird and Ainge were able to land General Manager roles shortly after their successful stints, while Kerr is a new-comer into the dataset. As such, the fixed effect estimate are expected to gain accuracy with longer tenure durations.

Table 7 compares a ranking of NBA head coaches by estimated fixed effective against a naïve approach consisting of career win percentages. Although some coaches figure at the top of

both ranking approaches, many appear to be overlooked by standard win records. A case-in-point is Brad Stevens who is 53rd by win percentage and 8th in estimated fixed effect. Conversely, Mike Brown is 9th by win percentage and 42nd by estimated ability, which suggests that his success could have been aided by superior talent (LeBron James). This is testament that team performance alone does not accurately capture the coach's contribution. While Stevens was able to elevate his roster to new heights, Brown's case was aided by superior talent .

It is important to note that using previous year's BPM as a proxy for coach-neutral ability level may exaggerate the effects of coaches joining young teams with budding superstars (Jason Kidd, Tom Thibodeau, Vinny Del Negro). Since most players have a negative BPM early in their career, this may influence the results. Another source of bias could be first-year players (i.e. rookies), who do not possess any prior metrics. In this analysis, all rookies are assumed to be on par with league average.

The range of the estimated fixed effects is quite large, with a difference of more than 30 wins per season between the coaches at the top and those at the bottom of the ranking (Figure 3). This suggests that NBA head coaches are heterogenous in their abilities.

Overall, these results are in-line with findings from Bertrand and Schoar (2003) [6], Graham et al. (2012) [13], and Muehlheusser et al. (2016) [15], which suggest that managers play a deterministic role in organizational success. Therefore, the implications for general managers and executives running basketball teams are quite sizeable. While the focus seems to lie on players, much can be achieved by undertaking changes at the top. Once again, the magnitude of these effects are large enough to make the difference between a mediocre and a playoff team but also between a playoff and a championship team.

4 Robustness

This section includes two robustness checks. First, the out-of-sample predictive power of the fixed effects is tested using cross validation. Second, the robustness of the results is tested against different tenure length considerations. Since longer tenures may be susceptible to endogeneity

between the coaching variable and performance metrics, this step explores the strength of the results to shorter tenures by imposing limits on coach-team pairs.

4.1 Leave One Out Cross-Validation

To explore whether the estimated fixed effects have out-of-sample predictive power, I consider a leave-one-out cross validation approach. This method consists of iteratively re-estimating the model each time leaving one team out of the equation. The resulting estimates are subsequently tested for statistical significance in explaining the performance of that very team that is left out.

This is also known as "jackknife" approach (Angrist, Imbens, and Krueger, 1999) and serves to determine whether a coach's all except one tenure provide useful information about the remaining tenure. For example, to infer whether Pat Riley's fixed effect is useful to predict the performance of the Miami Heat under his leadership, the fixed effect that is applied is based only upon his contribution to the L.A. Lakers and the New York Knicks. Similarly, only his fixed effect from his time on the Knicks and the Heat is tested to explain the Lakers' performance. As such, no information from a team's own historical perfromance is applied to that very same team, making it an out-of-sample exercise.

Table 3 displays the results, where Model I includes the roster quality variable alone, and Model II adds the coach fixed effects estimated using the leave-one-out approach. The findings show that both variables are useful in predicting team performance. While the measure of roster ability is associated with statistically significant coefficients in both models, adding the measure of coaching ability improves the model's adjusted R-squared by about half a percentage point. More importantly, the coefficient on coach fixed effects is statistically significant at the 1 percent level. It can be therefore inferred that the estimates of coaching ability developed in this paper have substantial use in predicting performance for newly assigned coach-team pairs.

Table 3: Cross-Validation: Predicting Out-of-sample Performance

	Dependent variable: $Wins$	
	Model I	Model II
$RosterQuality$	7.97***	6.94***
	(-0.26)	(-0.26)
$CoachFixedEffects$	-	.16***
		(-0.04)
$Constant$	33.4***	33.3***
	(-0.51)	(-0.53)
N	866	866
Adj. R-squared (%)	43.13	43.67

Asterisks indicate statistical significance at the 10 percent (*), 5 percent (**), and 1 percent (***) levels.
In *Model II*, coach fixed effects are estimated using the leave-one-out approach. Time period: 1978-79 until 2016-17 season. Note: Standard errors in parentheses are clustered at the team level.

4.2 Addressing the Use of Performance Metrics

Since the main model employs performance-based metrics to capture coach-neutral variations across teams, there is a concern that performance in previous seasons could be influenced by the coach in case he was already in charge. In this section, I explore the robustness of the fixed effects to shorter tenure considerations, where the endogeneity from using performance metrics is reduced to a large degree. For example, Model A (Table 4) imposes a maximum of four seasons per coach-team pair, where all subsequent seasons, during which one suspects that the endogeneity increases, are eliminated. Similarly, Models B and C impose three and two observations per coach-team tenures.

The key question is whether using lagged performance metrics as a roster quality measure biases the results. It must be noted that imposing these limits leads to sample reductions in Models A, B, and C. With lower number of observations per head coach, the standard errors are also subject to compromise. Aside from that, the estimated fixed effects from those models correlate quite well with those from the base model, as evidenced by the high spearman rank coefficients. Unsurprisingly, the correlation drops from 96.6 to 87.3 percent moving from 4 to 2 observations per coach-team pair, as more useful information is being omitted in shorter tenures. Overall, this suggests that the results are quite robust to alternative tenure considerations. In other words, the

potential endogeneity that could arise from longer coach-team tenures is not serious enough to question the model specification.

Table 4: Robustness to Different Tenure Length Limits

	Base Model		Model A		Model B		Model C	
	NoLimit		*4Seasons*		*3Seasons*		*2Seasons*	
Coach	*FE*	*Rk*	*FE*	*Rk*	*FE*	*Rk*	*FE*	*Rk*
Top Coaches								
Larry Bird	14.2	1	14.3	1	14.3	1	12.6	5
Steve Kerr	12.3	2	12.6	3	12.6	3	17.3	1
Gregg Popovich	11.7	3	9.5	10	11.7	5	9.3	17
Billy Cunningham	11.7	4	14.2	2	11.6	6	4.5	42
Stan Albeck	10.5	5	10.5	7	10.5	10	9.9	14
Scott Brooks	10.4	6	11.2	5	10.6	8	9.8	15
K.C. Jones	10.4	7	11.8	4	13.3	2	11.3	9
Brad Stevens	10	8	9.9	8	10.5	9	8.8	18
Tom Thibodeau	9.1	9	9.5	11	9.7	12	12.4	6
Danny Ainge	9.1	10	9.1	12	9.1	14	14.3	3
N			805		698		542	
Adjusted R-sq. (%)			57.3		58.7		62.3	
Spearman Rank Coef. (%)*			96.6		94		87.3	

Rank coefficients (Spearman) provide estimates of the strength of correlation with the full vector of fixed effect estimates from the base model. Models A, B, C restrict the dataset to 4, 3, and 2 observations per coach-team tenure, respectively. Note: FE = Fixed Effect.

5 Background Characteristics: Effect of Previous Playing Careers

Advanced skill at the underlying activity is viewed by some as a prerequisite for managerial success [12] (Goodall, Kahn, and Oswald, 2008). Examples of prominent managers who were once famous players include Larry Bird and Billy Cunningham. But evidently, not all excellent coaches were once excellent players. This section explores the effect of background characteristics by regressing the estimated coach fixed effects on background information, such as whether or not the coach was a professional player, and statistics from playing careers such assists and points per game, and career assist-to-turnover ratio.

The purpose is to determine whether there are common characteristics among coaches

who stand out in ability. Out of the sample of 165 coaches considered in this study, 86 are ex-professional NBA or ABA players , 47 occupied the Point Guard position , and 18 were selected as All-Stars at least once over the course of their careers. As a first step, I investigate whether having prior exposure as a player provides any systematic advantage. Then, focusing on those with playing career statistics available, I explore potential correlations between style of play and future coaching ability.

Table 5 displays the results for four models that use the estimated fixed effects as dependent variables. As explanatory variables, the first includes a dummy variable for whether the coach was a professional player earlier in his career. The second and third consider dummy variables for the Point Guard position and being an All-star caliber player. The remaining models regresses on a selection of box score statistics that could be correlated with coaching ability based on mainstream intuition, namely average points scored and average assists per game. The key question here is whether being particularly good at scoring or dishing out assists could be indicative of future coaching ability.

Table 5: Robustness to Different Tenure Length Limits

	Dependent Variable: $FixedEffectEstimate$				
	I	II	III	IV	V
Professional	-0.49	-0.33	-1.2		
	(-1.03)	(-1.26)	(-1.11)		
Point-Guard		-0.29			
		(-1.4)			
All-Star			3.42*		
			(-1.97)		
Points Per Game				0.06	
				(-0.12)	
Assists Per Game					0.14
					(-0.27)
Constant	0.25	0.26	0.25	-1.2	-0.99
	(-0.74)	(-0.74)	(-0.73)	(-1.95)	(-1.58)

Standard errors are reported in parentheses. Asterisks indicate statistical significance at the 10 percent (*), 5 percent (**), and 1 percent (***) level.

Overall, being an ex-professional player does not seem to matter. However, being an ex-All-star is partially correlated with high fixed effects. Perhaps having an extensive pedigree as a player serves a coach well in terms of influencing the performance of his players, having learned the ropes and excelled himself. Moreover, the hypothesis that point guards make for better coaches

does not reflect in the data. Finally, in terms of style play, the results provide little of anything statistically significant. It is important to note that, since the dependent variable is itself an estimated coefficient, the results of this exercise are subject to compounded standard errors and should therefore be considered as exploratory rather than conclusive.

These findings suggest that by no means should former players be favored in leading NBA teams. It is likely that the lack of managerial of experience may be at fault, since many tend to jump into a coaching role shortly after ending their playing career. On the other hand, coaches with no prior experience playing professionally typically go through a more difficult path onto becoming NBA head coaches. By the time they are promoted into the NBA, their managerial qualities are already proven in the lower tier leagues, and they are ready to contribute immediately.

6 Conclusion

The purpose of this paper is to shed light on the role of coaches and provide a quantitative dive into the relative size of their contribution. It accomplishes this objective by exploring the high frequency of front office changes in the NBA and disentangling their added-value while controlling for roster quality. The paper adds to the line of research on managerial contribution by being the first to apply the fixed effect approach in the context of the NBA. This method is highly utilized in scientific research, especially when analyzing the effect of manager identity on organizational performance. The basketball environment is particularly well-suited for this task since coaches are observed on multiple teams throughout their careers.

The findings provided in this paper demonstrate that, just as acquiring a star athlete like LeBron James can boost team wins, hiring the right manager can have the same effect. While having superior rosters can provide a head start, coaching ability is capable of elevating team performance to new heights. This can come through strategic game planning but also through leadership, motivation, and interpersonal skills. While some coaches excel at designing systems that maximize the relative skill of their players, others display an unmatched ability to derive value through building close relationships with players. While those may take place outside the

lines, superior leadership skills typically translates into enhanced teamwork and more inspired performances on the court. Both of these channels make it possible that the same set of players achieves different results under different head coaches.

To analyze the results, this paper shows that the developed estimates of coaching ability are useful in predicting team performance for newly formed coach-team pairs. In addition, it explores the effect of coaches' backgrounds as ex-professional players and finds that, on average, little can be inferred about future coaching ability from a player's box score statistics.

Despite thorough robustness checks, the approach adopted in this paper is not without its shortcomings. First, the relatively low number of observations for some coaches may over- or under-estimate their average fixed effect. This issue would lessen as more historical data points become available for those who are relatively new. Second, influential factors other than coaching and player talent may be omitted, such as front office quality and training facilities. For the sake of the analysis, these factors are assumed to balance out across the league. Finally, as mentioned previously, the use of lagged performance metrics as a measure of roster quality may exaggerate the effects of coaches who join teams with young and budding superstars.

Overall, the framework advanced in this paper should be of value to general managers and executives in positions to make hiring decisions. While much analytical information is available on players, little quantitative assessments are provided on coaches. This renders the task of identifying promising candidates particularly challenging. As discussed in this paper, relying at past winning records is misleading as many coaches seem to benefit from superior talent levels. The fixed effect approach, meanwhile, succeeds in capturing the effect of roster quality and therefore offers a more reliable alternative for the assessment of coaches.

References

[1] John M. Abowd, Francis Kramarz, and David N. Margolis. High Wage Workers and High Wage Firms. *Econometrica*, 67(2):251–333, 1999.

[2] Richard Audas, Stephen Dobsan, and John Goddard. Team Performance and Managerial Change in the English Football League. *Economic Affairs*, 17(3):30–36, 1997.

[3] Paul Beaudry and Tim Willems. On the macroeconomic consequences of over-optimism. Working Paper 24685, National Bureau of Economic Research, June 2018.

[4] David Berri, Michael Leeds, Eva Leeds, and Mike Mondello. The role of managers in team performance. *International Journal of Sport Finance*, 4:75–93, 05 2009.

[5] Christopher R. Berry and Anthony Fowler. How Much Do Coaches Matter. *MIT Sloan Sports Analytics Conference*, 2019.

[6] Marianne Bertrand and Antoinette Schoar. Managing with Style: The Effect of Managers on Firm Policies. *The Quarterly Journal of Economics*, 118(4):1169–1208, 2003.

[7] Albert A. Cannella and W. Glenn Rowe. Leader capabilities, succession, and competitive context: A study of professional baseball teams. *Leadership Quarterly*, 6(1):69–88, 1995.

[8] David Fabianic. Managerial change and organizational effectiveness in major league baseball: Findings for the eighties. *Journal of Sport Behavior*, 17(3):135–147, 1994.

[9] William A. Gamson and Norman A. Scotch. Scapegoating in Baseball. *American Journal of Sociology*, 70(1):69–72, 1964.

[10] Robert C. Giambatista. Jumping through hoops: A longitudinal study of leader life cycles in the NBA. *The Leadership Quarterly*, 15(5):607–624, 2004.

[11] Daniel Goleman. Leadership that gets results. *MIT Sloan Sports Analytics Conference*, 2000.

[12] Amanda H. Goodall, Lawrence M. Kahn, and Andrew J. Oswald. Why do leaders matter? A study of expert knowledge in a superstar setting. *Journal of Economic Behavior and Organization*, 77(3):265–284, 2011.

[13] John R. Graham, Si Li, and Jiaping Qiu. Managerial Attributes and Executive Compensation. *The Review of Financial Studies*, 25(1):144–186, 2012.

[14] Edward P. Lazear, Kathryn L. Shaw, and Christopher T. Stanton. The Value of Bosses. *Journal of Labor Economics*, 33(4):823–861, 2015.

[15] Gerd Muehlheusser, Sandra Schneemann, Dirk Sliwka, and Niklas Wallmeier. The Contribution of Managers to Organizational Success: Evidence from German Soccer. *Journal of Sports Economics*, 19(6):786–819, 2016.

[16] Dennis Smart, Jason Winfree, and Richard Wolfe. Major League Baseball Managers: Do They Matter? *Journal of Sport Management*, 22(3):303–321, 2008.

A Appendix

Table 6: Ranking of NBA Coaches by Estimated FE vs. Win Percentage

	FixedEffect Approach				*Naive* Approach	
Rank	*Coach*	*FixedEffect*		*Rank*	*Coach*	*WinPct*
1	Larry Bird	14.20	***	1	Steve Kerr	0.84
2	Steve Kerr	12.26	***	2	Billy Cunningham	0.70
3	Gregg Popovich	11.72	***	3	Phil Jackson	0.70
4	Billy Cunningham	11.70	***	4	Larry Bird	0.69
5	Stan Albeck	10.52	***	5	Gregg Popovich	0.69
6	Scott Brooks	10.42	***	6	K.C. Jones	0.64
7	K.C. Jones	10.41	***	7	Pat Riley	0.64
8	Brad Stevens	9.98	***	8	Scott Brooks	0.62
9	Tom Thibodeau	9.09	**	9	Mike Brown	0.62
10	Danny Ainge	9.07	***	10	Erik Spoelstra	0.61
11	Phil Jackson	8.84	***	11	Tom Thibodeau	0.60
12	Jeff Van Gundy	8.84	***	12	Danny Ainge	0.60
13	Jason Kidd	8.49	***	13	Jerry Sloan	0.60
14	Stan Van Gundy	8.25	***	14	Stan Van Gundy	0.59
15	Chuck Daly	8.10	***	15	Chuck Daly	0.59
16	Vinny Del Negro	7.83	***	16	George Karl	0.59
17	Rick Carlisle	7.80	***	17	Rick Adelman	0.58
18	Pat Riley	7.61	***	18	Mike Budenholzer	0.58
19	Slick Leonard	7.04	***	19	Doc Rivers	0.58
20	Rick Adelman	6.51	***	20	Avery Johnson	0.58
21	Rick Pitino	6.43	***	21	Jeff Van Gundy	0.57
22	Erik Spoelstra	6.41	***	22	Rick Carlisle	0.57
23	Cotton Fitzsimmons	6.31	***	23	Larry Brown	0.57
24	Eric Musselman	6.19	**	24	Don Nelson	0.56
25	Don Nelson	5.99	***	25	Kevin McHale	0.56
26	Mike D'Antoni	5.99	*	26	Rudy Tomjanovich	0.56
27	Frank Vogel	5.98	**	27	Dave Joerger	0.55
28	Jack Ramsay	5.92	***	28	Mike Fratello	0.55
29	Allan Bristow	5.84	***	29	Del Harris	0.55
30	Kevin McHale	5.83		30	Frank Vogel	0.54
31	Mike Budenholzer	5.83	***	31	Lenny Wilkens	0.54
32	Frank Layden	5.49	***	32	Red Holzman	0.54
33	Nate McMillan	5.45	***	33	Doug Moe	0.54
34	Mark Jackson	5.42	***	34	Bill Russell	0.54
35	Larry Brown	5.06	***	35	Vinny Del Negro	0.53
36	Flip Saunders	5.02	**	36	Mike D'Antoni	0.53
37	Doug Collins	4.97	***	37	Mark Jackson	0.53
38	George Karl	4.91	**	38	Dwane Casey	0.53
39	Dave Joerger	4.80	***	39	Paul Westphal	0.53
40	Dwane Casey	4.55		40	Mike Schuler	0.53
41	Steve Clifford	4.53	***	41	Stan Albeck	0.52
42	Mike Brown	4.53	***	42	Slick Leonard	0.52
43	Quin Snyder	4.49	***	43	Cotton Fitzsimmons	0.52
44	Monty Williams	4.49	***	44	Jack Ramsay	0.52
45	Bill Fitch	4.42		45	Flip Saunders	0.52
46	Paul Westphal	4.08		46	Doug Collins	0.52
47	Mike Fratello	4.07		47	Quin Snyder	0.52
48	Tyrone Corbin	4.05	***	48	John MacLoed	0.52
49	Gene Shue	4.04	***	49	Al Attles	0.52
50	Lenny Wilkens	3.99	*	50	Brad Stevens	0.51
51	Red Holzman	3.89	***	51	Nate McMillan	0.51
52	Doc Rivers	3.79	**	52	Bob Hill	0.51
53	Avery Johnson	3.77		53	Bob Bass	0.51
54	Lionel Hollins	3.64		54	Allan Bristow	0.50
55	Isiah Thomas	3.42		55	Scott Skiles	0.50
56	Jerry Sloan	3.26	**	56	Jason Kidd	0.49
57	Rudy Tomjanovich	3.12	**	57	Frank Layden	0.49
58	John MacLoed	2.93	**	58	Steve Clifford	0.49
59	P.J. Carlesimo	2.72		59	Lionel Hollins	0.49
60	Jack McKinney	2.65	***	60	Brian Hill	0.49
61	Jimmy Rodgers	2.53		61	Hubie Brown	0.49
62	Del Harris	2.44		62	Terry Stotts	0.49
63	Mike Malone	2.11		63	Maurice Cheeks	0.49
64	Sam Mitchell	1.91	***	64	Gene Shue	0.48
65	Kevin Loughery	1.82		65	Jim O'Brien	0.48
66	Paul Westhead	1.75		66	Doug Motta	0.48

67	Larry Drew	1.38		67	Rick Pitino	0.47
68	Mike Woodson	1.33		68	Bill Fitch	0.46
69	Paul Silas	1.30		69	Isiah Thomas	0.46
70	Brian Hill	1.27		70	Kevin Loughery	0.46
71	Hubie Brown	1.16	*	71	Larry Drew	0.46
72	Doug Moe	0.73		72	Mike Woodson	0.46
73	Jim O'Brien	0.45		73	Mike Dunleavy	0.46
74	Tom Nissalke	0.44		74	Dave Cowens	0.46
75	Bob Weiss	0.36		75	Chris Ford	0.46
76	Doug Motta	0.25		76	Alvin Gentry	0.46
77	Mike Dunleavy	0.08		77	Jim Lynam	0.46
78	Al Attles	-0.28		78	Dan Issel	0.46
79	Terry Stotts	-0.45		79	Paul Westhead	0.45
80	Jeff Hornacek	-0.46		80	Jeff Hornacek	0.45
81	Bob Hill	-0.53		81	Lawrence Frank	0.45
82	Dave Cowens	-0.84		82	Eric Musselman	0.44
83	Chris Ford	-1.10		83	Monty Williams	0.44
84	Keith Smart	-1.13		84	Paul Silas	0.44
85	Eddie Jordan	-1.24		85	Bernie Bickerstaff	0.44
86	Scott Skiles	-1.56		86	P.J. Carlesimo	0.43
87	Bernie Bickerstaff	-1.82		87	Sam Mitchell	0.43
88	Johnny Batch	-1.83	***	88	Bob Weiss	0.43
89	Randy Wittman	-1.86		89	Eddie Jordan	0.43
90	Mike Schuler	-2.19		90	Matt Guokas	0.43
91	Alvin Gentry	-2.24	**	91	Phil Johnson	0.43
92	Matt Guokas	-2.31		92	Tyrone Corbin	0.42
93	Wes Unseld	-2.56	***	93	Jimmy Rodgers	0.42
94	Jim Lynam	-2.61		94	Tom Nissalke	0.42
95	Don Chaney	-2.73		95	Mike Malone	0.41
96	Ron Rothstein	-2.74		96	Randy Wittman	0.41
97	Garry St. Jean	-2.75		97	Don Chaney	0.41
98	George Irvine	-3.42		98	Byron Scott	0.41
99	Byron Scott	-3.43	***	99	John Lucas	0.40
100	John Lucas	-3.43		100	Jack McKinney	0.39
101	Lawrence Frank	-3.92	***	101	Garry St. Jean	0.38
102	Dan Issel	-4.17	***	102	Scotty Robertson	0.38
103	Bill Musselman	-4.61	*	103	Wes Unseld	0.37
104	Maurice Cheeks	-4.71		104	Johnny Batch	0.36
105	Phil Johnson	-5.11	***	105	Don Casey	0.36
106	Scotty Robertson	-6.79		106	Keith Smart	0.35
107	Brett Brown	-7.17	***	107	Richie Adubato	0.35
108	Don Casey	-8.37	***	108	George Irvine	0.34
109	Richie Adubato	-9.61	***	109	Ron Rothstein	0.30
110	Tim Floyd	-9.93	***	110	Bill Musselman	0.30
111	Kurt Rambis	-10.02		111	Tim Floyd	0.28
112	Sydney Lowe	-11.97	***	112	Kurt Rambis	0.28
113	Bill Russell	-14.84	***	113	Sydney Lowe	0.26
114	Bob Bass	-17.88	***	114	Brett Brown	0.23

Asterisks indicate statistical significance at the 10 percent (*), 5 percent (**), and 1 percent (***) level. Time Period 1978-79 until 2016-17 seasons. Note: Only showing those with at least 3 years of coaching experience.

WinPct = Career win percentages until 2016-17 season (regular seasons only).

Table 7: Descriptive Statistics - Full Sample

	Name	Seasons	Teams		Name	Seasons	Teams
1	Adelman, Rick	22	5	83	Kerr, Steve	3	1
2	Adubato, Richie	4	3	84	Kidd, Jason	4	2
3	Ainge, Danny	3	1	85	Kruger, Lon	2	1
4	Albeck, Stan	8	5	86	Krystkowiak, Larry	1	1
5	Atkinson, Kenny	1	1	87	Kuester, John	2	1
6	Attles, Al	13	1	88	Layden, Frank	7	1
7	Bass, Bob	7	5	89	Leonard, Slick	14	4
8	Batch, Johnny	3	1	90	Littles, Gene	2	3
9	Beard, Butch	2	1	91	Loughery, Kevin	17	8
10	Bickerstaff, Bernie	11	5	92	Lowe, Sydney	4	2
11	Bickerstaff, J.B.	1	1	93	Lucas, John	5	3
12	Bird, Larry	3	1	94	Lue, Tyronn	2	1
13	Blair, Bill	1	2	95	Lynam, Jim	9	3
14	Blatt, David	2	1	96	MacLoed, John	17	3
15	Bristow, Allan	5	1	97	Malone, Brendan	1	2
16	Brooks, Scott	8	2	98	Malone, Mike	3	2
17	Brown, Brett	4	1	99	McHale, Kevin	5	2

18	Brown, Hubie	13	4
19	Brown, Larry	28	11
20	Brown, Mike	7	2
21	Buckner, Quinn	1	1
22	Budenholzer, Mike	4	1
23	Bzdelik, Jeff	2	1
24	Calipari, John	2	1
25	Carlesimo, P.J.	7	4
26	Carlisle, Rick	15	3
27	Carr, M.L.	2	1
28	Carter, Butch	2	1
29	Carter, Fred	1	1
30	Cartwright, Bill	2	1
31	Casey, Don	3	2
32	Casey, Dwane	7	2
33	Chaney, Don	10	4
34	Cheeks, Maurice	8	3
35	Cleamons, Jim	1	1
36	Clifford, Steve	4	1
37	Collins, Doug	11	4
38	Corbin, Tyrone	4	2
39	Cowens, Dave	4	3
40	Cunningham, Billy	8	1
41	Curry, Michael	1	1
42	D'Antoni, Mike	12	5
43	Daly, Chuck	14	4
44	Davis, Johnny	3	3
45	Del Negro, Vinny	5	2
46	Donavon, Billy	2	1
47	Drew, Larry	4	2
48	Dunlap, Mike	1	1
49	Dunleavy, Mike	17	4
50	Fisher, Derek	2	1
51	Fitch, Bill	25	5
52	Fitzsimmons, Cotton	20	5
53	Fizdale, David	1	1
54	Floyd, Tim	4	2
55	Ford, Chris	9	4
56	Frank, Lawrence	8	2
57	Fratello, Mike	15	3
58	Gentry, Alvin	11	5
59	Guokas, Matt	7	2
60	Hamilton, Leonard	1	1
61	Hanzlick, Bill	1	1
62	Harris, Del	12	3
63	Harter, Dick	1	1
64	Hill, Bob	7	4
65	Hill, Brian	8	2
66	Hoiberg, Fred	2	1
67	Hollins, Lionel	7	2
68	Holzman, Red	16	3
69	Hornacek, Jeff	4	2
70	Iavaroni, Marc	2	1
71	Irvine, George	4	2
72	Issel, Dan	5	1
73	Jackson, Mark	3	1
74	Jackson, Phil	20	2
75	Jackson, Stu	2	2
76	Joerger, Dave	4	2
77	Johnson, Avery	6	2
78	Johnson, Frank	2	1
79	Johnson, Phil	7	3
80	Jones, K.C.	10	5
81	Jordan, Eddie	7	3
82	Karl, George	25	6
100	McKinney, Jack	4	3
101	McMillan, Nate	12	3
102	Mitchell, Sam	5	2
103	Moe, Doug	14	3
104	Montgomery, Mike	2	1
105	Motta, Doug	24	5
106	Musselman, Bill	3	3
107	Musselman, Eric	3	2
108	Nelson, Don	30	4
109	Nissalke, Tom	10	7
110	O'Brien, Jim	8	3
111	O'Neill, Kevin	1	1
112	Pfund, Randy	2	1
113	Pitino, Rick	5	2
114	Popovich, Gregg	21	1
115	Porter, Terry	3	2
116	Rambis, Kurt	3	3
117	Ramsay, Jack	20	4
118	Reed, Willis	3	2
119	Reynolds, Jerry	2	1
120	Riley, Pat	24	3
121	Rivers, Doc	17	3
122	Robertson, Scotty	4	3
123	Rodgers, Jimmy	3	2
124	Rothstein, Ron	4	2
125	Russell, Bill	8	3
126	Saunders, Flip	16	3
127	Schuler, Mike	4	2
128	Scott, Byron	14	4
129	Shaw, Brian	2	1
130	Shue, Gene	20	4
131	Silas, Paul	11	4
132	Skiles, Scott	12	4
133	Sloan, Jerry	25	2
134	Smart, Keith	3	3
135	Snyder, Quin	3	1
136	Spoelstra, Erik	9	1
137	St. Jean, Garry	5	2
138	Stevens, Brad	4	1
139	Stotts, Terry	8	3
140	Theus, Reggie	1	1
141	Thibodeau, Tom	6	2
142	Thomas, Isiah	5	2
143	Tomjanovich, Rudy	12	2
144	Triano, Jay	3	1
145	Unseld, Wes	7	1
146	Van Gundy, Jeff	10	2
147	Van Gundy, Stan	10	3
148	Vaughn, Jacque	3	1
149	Versace, Dick	2	1
150	Vincent, Sam	1	1
151	Vogel, Frank	6	2
152	Walker, Darrell	2	2
153	Walsh, Donnie	2	1
154	Walton, Luke	1	1
155	Watson, Earl	1	1
156	Weiss, Bob	6	4
157	Westhead, Paul	5	3
158	Westphal, Paul	8	3
159	Wetzel, John	1	1
160	Wilkens, Lenny	31	6
161	Williams, Monty	5	1
162	Winters, Brian	2	2
163	Wittman, Randy	8	3
164	Wohl, Dave	2	1
165	Woodson, Mike	8	2

Time Period 1978-79 until 2016-17 seasons.